

## ANDRAVIDA, Greece

WMO: 166820

Lat: 37.923N

Lon: 21.288E

Elev: 12

StdP: 101.18

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |            | Humidification DP/MCDB and HR |            |            |             |            |            | Coldest Month WS/MCDB |             |            |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|------------|------------|-------------------------------|------------|------------|-------------|------------|------------|-----------------------|-------------|------------|-------------|-----------------------|------------|--------------|
|               |            |            | 99.6%                         |            |            | 99%         |            |            | 0.4%                  |             | 1%         |             |                       |            |              |
|               | 99.6%      | 99%        | DP                            | HR         | MCDB       | DP          | HR         | MCDB       | WS                    | MCDB        | WS         | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>0.3 | (c)<br>2.0 | (d)<br>-5.0                   | (e)<br>2.5 | (f)<br>4.4 | (g)<br>-3.0 | (h)<br>2.9 | (i)<br>5.0 | (j)<br>11.3           | (k)<br>13.1 | (l)<br>9.8 | (m)<br>13.1 | (n)<br>0.7            | (o)<br>100 | (p)<br>0.389 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 12.0                   | 34.0            | 21.6 | 32.9 | 21.7 | 31.7 | 21.7 | 24.5                | 29.5 | 23.9 | 29.1 | 23.3 | 28.6 | 4.4                  | 300  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 23.1                            | 17.9 | 26.9 | 22.2 | 16.9 | 26.5 | 21.6 | 16.3 | 26.2 | 74.3          | 29.5 | 71.7 | 28.9 | 69.3 | 28.9 | 27.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 9.1               | 7.7  | 6.5 | -2.5                       | 36.7 | 1.9                | 1.7 | -3.9                                               | 37.9 | -5.1       | 38.9 | -6.2       | 39.9 | -7.6       | 41.1 |
|                   |      |     | -3.5                       | 25.7 | 1.8                | 0.9 | -4.8                                               | 26.3 | -5.8       | 26.8 | -6.8       | 27.3 | -8.1       | 27.9 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 17.3   | 9.7   | 10.2  | 12.0  | 14.8  | 19.1  | 22.9  | 25.1  | 25.7  | 22.7  | 18.7  | 14.7  | 11.1 |
|                                                                                  | DBStd       | 6.15   | 2.86  | 2.81  | 2.43  | 2.42  | 2.26  | 2.25  | 1.65  | 1.53  | 2.11  | 2.51  | 2.86  | 2.99 |
|                                                                                  | HDD10.0     | 104    | 39    | 28    | 9     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 3     | 23   |
|                                                                                  | HDD18.3     | 1178   | 266   | 227   | 197   | 108   | 18    | 1     | 0     | 0     | 1     | 26    | 111   | 223  |
|                                                                                  | CDD10.0     | 2761   | 31    | 36    | 71    | 146   | 282   | 388   | 467   | 486   | 381   | 270   | 145   | 58   |
|                                                                                  | CDD18.3     | 793    | 0     | 0     | 1     | 2     | 42    | 139   | 208   | 228   | 132   | 37    | 3     | 1    |
|                                                                                  | CDH23.3     | 7699   | 0     | 0     | 4     | 37    | 341   | 1329  | 2221  | 2471  | 1072  | 215   | 10    | 0    |
|                                                                                  | CDH26.7     | 2922   | 0     | 0     | 1     | 3     | 60    | 464   | 931   | 1134  | 308   | 19    | 1     | 0    |
| Wind                                                                             | WSAvg       | 2.2    | 2.3   | 2.5   | 2.6   | 2.3   | 2.2   | 2.2   | 2.2   | 2.1   | 2.0   | 1.8   | 2.2   | 2.2  |
| Precipitation                                                                    | PrecAvg     | 750    | 104   | 90    | 62    | 39    | 23    | 7     | 2     | 8     | 50    | 91    | 138   | 138  |
|                                                                                  | PrecMax     | 946    | 284   | 260   | 151   | 110   | 67    | 48    | 20    | 74    | 108   | 200   | 294   | 330  |
|                                                                                  | PrecMin     | 414    | 10    | 9     | 0     | 4     | 1     | 0     | 0     | 0     | 1     | 9     | 2     | 0    |
|                                                                                  | PrecStd     | 134    | 59    | 61    | 37    | 25    | 15    | 10    | 5     | 15    | 29    | 54    | 67    | 70   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 18.2  | 20.0  | 22.2  | 26.9  | 30.2  | 34.8  | 35.2  | 36.8  | 33.0  | 28.9  | 24.7  | 20.2 |
|                                                                                  |             | MCWB   | 13.9  | 13.0  | 13.6  | 15.9  | 18.9  | 21.5  | 21.8  | 21.1  | 20.1  | 19.7  | 18.0  | 15.9 |
|                                                                                  | 2%          | DB     | 17.0  | 17.8  | 19.9  | 24.1  | 28.2  | 32.4  | 33.3  | 34.2  | 31.0  | 26.9  | 22.8  | 18.8 |
|                                                                                  |             | MCWB   | 13.6  | 13.2  | 13.8  | 15.4  | 18.5  | 21.5  | 22.1  | 21.6  | 21.0  | 19.2  | 17.4  | 14.9 |
|                                                                                  | 5%          | DB     | 15.9  | 16.3  | 18.2  | 22.1  | 26.8  | 30.9  | 32.2  | 33.1  | 29.9  | 25.7  | 21.2  | 17.2 |
|                                                                                  |             | MCWB   | 12.7  | 12.5  | 13.5  | 15.2  | 18.2  | 21.2  | 22.2  | 21.6  | 21.0  | 18.8  | 16.6  | 13.8 |
|                                                                                  | 10%         | DB     | 14.9  | 15.2  | 17.1  | 20.7  | 25.1  | 29.2  | 31.0  | 31.9  | 28.5  | 24.2  | 20.0  | 16.1 |
|                                                                                  |             | MCWB   | 11.8  | 11.8  | 13.1  | 14.8  | 17.8  | 20.9  | 22.1  | 21.8  | 20.8  | 18.3  | 15.8  | 13.2 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 15.2  | 15.1  | 15.7  | 17.9  | 21.2  | 24.4  | 25.2  | 25.6  | 24.5  | 22.2  | 19.6  | 16.7 |
|                                                                                  |             | MCDB   | 17.1  | 17.8  | 20.0  | 23.2  | 26.5  | 29.7  | 30.5  | 30.2  | 28.3  | 25.3  | 22.6  | 19.3 |
|                                                                                  | 2%          | WB     | 14.1  | 13.9  | 14.9  | 16.8  | 20.2  | 23.2  | 24.3  | 24.7  | 23.4  | 21.1  | 18.3  | 15.4 |
|                                                                                  |             | MCDB   | 16.4  | 16.5  | 18.4  | 21.9  | 25.2  | 29.1  | 29.7  | 29.6  | 27.8  | 24.5  | 21.5  | 17.8 |
|                                                                                  | 5%          | WB     | 12.9  | 13.1  | 14.1  | 16.0  | 19.3  | 22.5  | 23.7  | 24.0  | 22.5  | 20.1  | 17.3  | 14.6 |
|                                                                                  |             | MCDB   | 15.3  | 15.9  | 17.3  | 20.4  | 24.4  | 28.3  | 29.1  | 29.1  | 27.2  | 24.1  | 20.4  | 16.9 |
|                                                                                  | 10%         | WB     | 12.0  | 12.3  | 13.4  | 15.2  | 18.6  | 21.7  | 23.1  | 23.4  | 21.7  | 19.3  | 16.4  | 13.4 |
|                                                                                  |             | MCDB   | 14.3  | 15.1  | 16.5  | 19.3  | 23.6  | 27.4  | 28.6  | 28.6  | 26.5  | 23.1  | 19.4  | 15.8 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 8.4   | 8.7   | 9.4   | 10.3  | 11.1  | 11.6  | 11.7  | 12.0  | 10.7  | 10.4  | 9.3   | 8.4  |
|                                                                                  |             | MCDBR  | 8.8   | 9.5   | 11.0  | 13.1  | 13.6  | 14.0  | 13.5  | 14.4  | 12.4  | 11.6  | 10.3  | 8.9  |
|                                                                                  |             | MCWBR  | 6.1   | 6.1   | 6.6   | 6.4   | 5.8   | 5.9   | 5.6   | 5.7   | 5.6   | 5.8   | 6.1   | 6.0  |
|                                                                                  | 5% WB       | MCDBR  | 7.9   | 8.7   | 10.0  | 11.3  | 11.8  | 12.6  | 11.9  | 11.9  | 11.0  | 9.8   | 9.2   | 7.4  |
|                                                                                  |             | MCWBR  | 5.9   | 6.0   | 6.5   | 6.3   | 5.8   | 5.9   | 5.6   | 5.6   | 5.7   | 5.7   | 6.1   | 5.5  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.353  | 0.383 | 0.424 | 0.464 | 0.456 | 0.434 | 0.439 | 0.449 | 0.432 | 0.410 | 0.381 | 0.361 |      |
|                                                                                  | taud        | 2.434  | 2.321 | 2.195 | 2.099 | 2.183 | 2.297 | 2.287 | 2.256 | 2.301 | 2.331 | 2.382 | 2.412 |      |
|                                                                                  | Ebn at Noon | 820    | 836   | 832   | 819   | 832   | 849   | 841   | 824   | 816   | 795   | 777   | 779   |      |
|                                                                                  | Edh at Noon | 86     | 108   | 134   | 157   | 147   | 131   | 131   | 132   | 120   | 106   | 90    | 82    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 2.08   | 2.92  | 4.31  | 5.64  | 6.73  | 7.63  | 7.83  | 6.96  | 5.25  | 3.68  | 2.48  | 1.89  |      |
|                                                                                  | RadStd      | 0.23   | 0.29  | 0.32  | 0.44  | 0.32  | 0.35  | 0.22  | 0.21  | 0.28  | 0.27  | 0.21  | 0.24  |      |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                        | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) | Station Only           | +0.24 | N/A     | N/A    | +0.78   | -0.48 | -0.82 | N/A         | -67     | +100    | N/A     |
| (47) | Regional (8 neighbors) | +0.30 | N/A     | N/A    | +0.78   | N/A   | N/A   | N/A         | -76     | +105    | N/A     |

Nomenclature: See separate page